The Horizontal Geometry of Production Networks

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Background

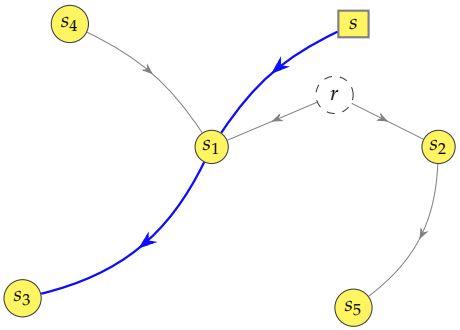
- Comovement across sectors is an essential feature of business cycles {*e.g.*, Lucas (1977)}
  - driven by independent micro-level shocks, not aggregate {*e.g.*, Gabaix (2011), Foerster *et al.* (2011)}

- Production networks $\longrightarrow$ role of Input-Output linkages {*e.g.*, Long and Plosser (1983)}
  - an idiosyncratic shock propagates to others along supply chains $\leftrightarrow$ "cascade" effects;
  - economically dominant sectors generate aggregate fluctuations {*e.g.*, Acemoglu *et al.* (2012)}.

→ Channel: ***vertical complementarities***, through network trade intensities {*e.g.*, Shea (2002)}
  - *linear economies* $\leftrightarrow$ comovement from even propagation of shocks {most of the literature};
  - *non-linear complementarities in production* $\leftrightarrow$ the even transmission of shocks is reversed, thereby allowing opposite comovement {*e.g.*, Baqaee and Farhi (2019)}.

Stylized propagation mechanisms

Verticality



This paper: central intuition

- Beside the *vertical* dimension, there exists an ***horizontal*** geometry of the network
  - ☞ sectors are additionally held together by *demand and supply interdependencies from having in common a similar Input-Output structure*
  - ✍ coexistence in the same primitive structure, not an additional assumption ▸ Representation

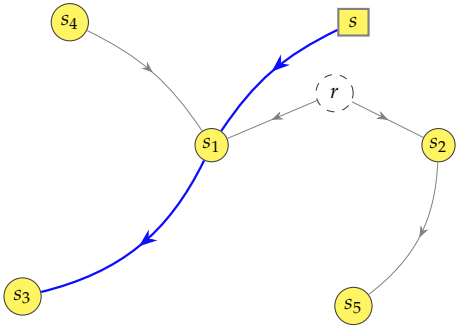
- Horizontality $\longrightarrow$ network "economic" distances between paired sectors
  - ***input demand*** distance, when *buying from* similar upstream sectors;
  - ***input supply*** distance, when *selling to* similar downstream sectors.

- These novel horizontal interdependencies add to the vertical transmission of shocks
  - network connections extend beyond trade flows $\leftrightarrow$ horizontal mapping

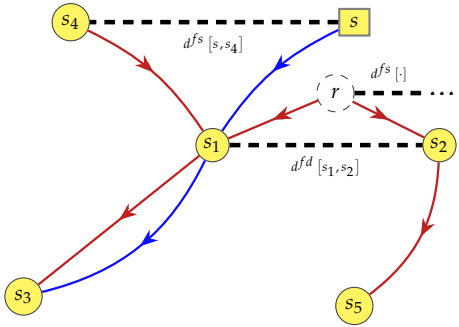
$\rightarrow$ "*who trades with whom*" + "*who depends on whom*"

Stylized propagation mechanisms

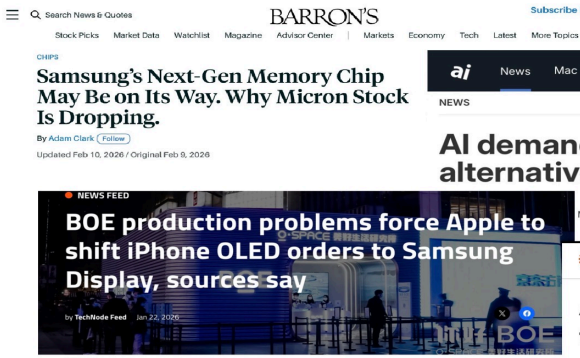
Verticality

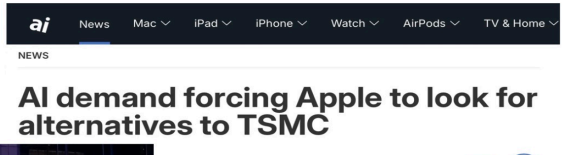


Verticality + Horizontality



Real world examples, not a "too theoretical" mechanism











▸ Supply: Micron ▸ Supply: AirPods ▸ Supply: OLED panels ▸ Demand: memory chips ▸ Demand: cloud ▸ Demand: TMSC

Central results

Question

how does the horizontal transmission impact comovement and aggregate dynamics?

Theory → ***sectoral comovement***: depends on the nature of the network "economic" distance

- ambiguous in *demand*, opposite in *supply*

→ ***aggregate dynamics***: endogenous response of Domar weights to shocks

- not cascading amplification, but widespread reallocation
- hidden in classical models {*e.g.*, Acemoglu *et al.* (2012)} due to the Hulten's (1978) theorem

Empirics → a ***Bartik-style estimation*** studies the transmission of aggregate shocks

- *horizontal* propagation transmits differently U.S. productivity and government spending

→ ***Local Projections*** on U.S. employment data ratify the insights on comovement

Structure of the paper

① Foundational theory, to set network distances and their characteristics (hints)

② Horizontal transmission, to shape aggregates from horizontal linkages (hints)

- inspect how shocks propagate and cumulate considering network distances

③ Mechanism at work, data-driven Input-Output estimation of

- vertical vs. horizontal propagating patterns;
- horizontal transmission upon aggregate shocks.

④ Empirical validation, to test comovement from sectoral variations (not today)

- network effects of nearby sectors' employment changes on sectoral employment

⑤ Policy implications, for future novel perspectives (not today)

- improving the design of policy interventions through the network's horizontal geometry

Foundational Theory

Distance-lead comovement

- Exploit the distance's definitions to relate sectoral optimality conditions → Covariances
  - on the input side, sectors buying from the same suppliers rebalance input costs
    - ⋆ *hard to interpret* due to substitution patterns across common upstream suppliers → Proposition 1
  - on the output side, sectors selling to the same buyers exhibit relative price changes
    - ⋆ *co-movement is transparent* due to simultaneous competition for downstream buyers → Proposition 2

- Systematic adjustments scaled by *distance matrices*, $\boldsymbol{\mathcal{D}}^j$ for $j = \{fd, fs\}$ → Interpretation

✍ **Lemma 1** formalizes their characteristics, such as: → Lemma 1
  - *non-negative* and *full*: all sectors are horizontally connected, even if not trading vertically;
  - a unique Input-Output structure has different network distance relations;
  - conditions under which distances are uniquely determined (demand = supply).

To what extent horizontality contributes to aggregates?

① **Theorem 1** → *Horizontal transmission moves Domar weights* ▸ Theorem 1

$$\underbrace{d\boldsymbol{\lambda}}_{\text{Domar}} = \underbrace{\boldsymbol{\mathcal{L}}'}_{\text{back to the supplier}} \underbrace{\boldsymbol{\mathcal{H}}}_{\text{downstream reallocation}} \underbrace{\boldsymbol{\mathcal{L}}}_{\text{cascade from the supplier}} \underbrace{d\log \mathbf{z}}_{\text{shock}}$$

- *vertical structure* → which commodities are affected (cascade + anti-cascade)
- *horizontal structure* → how downstream expenditure reallocates across common suppliers

- With Cobb-Douglas intermediates, horizontal transmission disappears ▸ Corollary 1
  - expenditure shares on inputs are fixed and invariant to the shocks, fixed Domar
- Horizontal effects cannot be recovered from the vertical structure alone ▸ Corollary 2
- Horizontal relations become increasingly important as the network expands ▸ Corollary 3

To what extent horizontality contributes to aggregates?

- In efficient economies, Hulten's (1978) theorem holds at every equilibrium

$$d \log Y = \boldsymbol{\lambda}' d \log \boldsymbol{z}$$

  - it fixes Domar weights → whatever the arrangement of the Input-Output relationships
  - *accounting role*: at first-order, horizontal geometry seems to vanish

→ Hulten gives the slope of aggregate output; Theorem 1 gives how this slope changes. Next?

② **Theorem 2** → *Horizontal geometry shifts the mean of aggregate GDP* ▸ Theorem 2

$$\underbrace{d^2 \log Y}_{\text{GDP}} = \underbrace{d \log \boldsymbol{z}'}_{\text{shock}} \underbrace{\boldsymbol{\mathcal{L}}'}_{\text{input-price anti-cascade}} \underbrace{\boldsymbol{\mathcal{H}}}_{\text{price dispersion}} \underbrace{\boldsymbol{\mathcal{L}}}_{\text{input-price cascade}} \underbrace{d \log \boldsymbol{z}}_{\text{shock}}$$

  - *second order effects* on curvature: horizontal reallocation generates aggregate gains/losses

- Consequently, it holds $\mathbb{E}\left[\log Y\right] - \log Y^{det} \neq 0$

→ differently from Acemoglu *et al.* (2012), even if Lucas's (1977) diversification holds the network's horizontal geometry generates aggregate dynamics

Empirics

Data

- From the United States' Bureau of Economic Analysis (BEA) I collect information, for 3-digit U.S. 2017 NAICS sectors, on *employment* and on: I-O data I-O structure
  - inter-sectoral linkages exploiting Input-Output (I-O) matrices:
    - ⋆ share of commodity production (*market share, after redefinitions* tables);
    - ⋆ I-O weights in producer values (*commodity-by-commodity total direct requirements* tables).
  - disregard small I-O transactions {*e.g.*, Acemoglu *et al.* (2012), Carvalho (2014)} to build distances
    - ⋆ no link if sector-s's total input sales/purchases are less than 1%
- *Unit*: 65 private sectors at 3-digit U.S. 2017 NAICS
- *Period*: 1998-2022
- *Network structure* at 2007 *robust at different years*

Network representation, sales Network representation, purchases

Setting the stage

- Degree of separation between sectors within the network (low value, low distance)
  - entries of distance matrices are *Euclidean distances*, so that $d^j[s, s'] \in [0, 1]$ ▸ Intensive margin

$$d^{fd}[i, j] = \sqrt{[\alpha_{ii} - \alpha_{ji}]^2 + [\alpha_{ij} - \alpha_{jj}]^2 + \cdots + [\alpha_{im} - \alpha_{jm}]^2}$$

    from the *Total Requirements* Input-Output network

  - then, I consider *extensive margins*, $d^q_{ext}[i, j] = \{1, 2, \ldots, d_{max}\}$ ▸ Graph ▸ Characteristics
    - ⋆ computed as a ***shortest path problem*** (how many steps to reach other nodes) ▸ Algorithm

- Before the empirics, sectoral propagation patterns using matrices' spectral norms
  - how important each sector is for the transmission of micro-level shocks
  - key *vertical* propagation hubs vs. widespread *horizontal* transmission

Propagating patterns

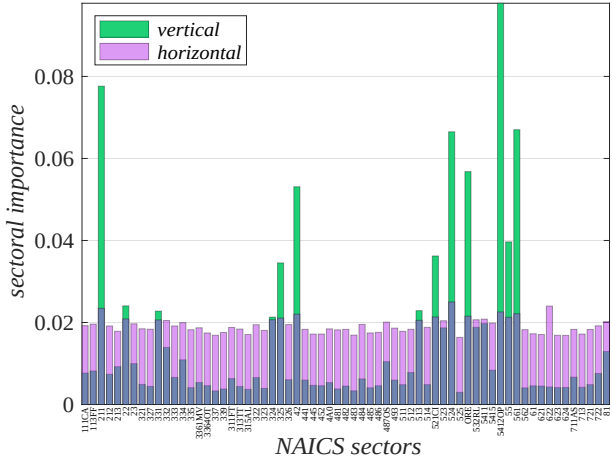


Figure 3: Sectoral influence in shock transmission

- Horizontal dominance

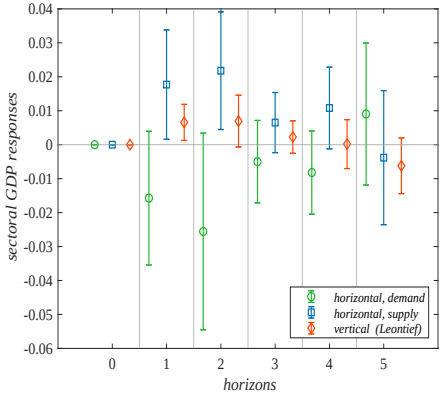
Validating horizontal geometries: a Bartik-style design

- So far, sector-level importance $\leftrightarrow$ now, aggregate shocks hit all simultaneously
  - *cross-sectional variation*: sectors differ in their ***network exposure*** to the same disturbance G_t
- Network structure interacting with predetermined sensitivities to aggregate shock
  - horizontal exposure ***residualised*** from orthogonal vertical (Leontief inverse) exposure, $\hat{r}^{\mathcal{D}^j}$
- ***Shift-Share Local Projection*** (LP-IV) estimation

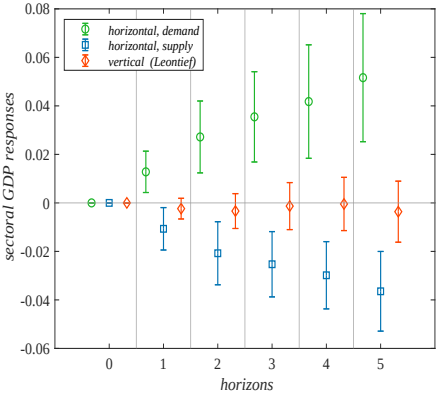
$$\begin{aligned}\Delta y_{t+h}(s) &= \beta_h^{\mathcal{D}^{fd}}\left(G_t \times \widehat{r}^{\mathcal{D}^{fd}}(s)\right) + \beta_h^{\mathcal{D}^{fs}}\left(G_t \times \widehat{r}^{\mathcal{D}^{fs}}(s)\right) \\ &\quad + \beta_h^{\mathcal{V}}\left(G_t \times exposure^{\mathcal{V}}(s)\right) + \phi(t+h) + \phi_h(s) + v_{t+h}^{y}(s)\end{aligned}$$

  - $\beta_h^{\mathcal{D}^j}$ measures *differential* sectoral responses across exposure levels, not unconditional
- Identification is purely cross-sectional
  - at each horizon-h, if $\beta_h^{\mathcal{D}^j} > 0 \Rightarrow$ amplification; if $\beta_h^{\mathcal{D}^j} < 0 \Rightarrow$ attenuation via the network

Network exposures and aggregate shocks



(**a**) Total Factor Productivity



(**b**) Government spending

Figure 4: "Networking up" aggregate shocks

Interpretation

Conclusion

Take-away remarks

- ❉ *Horizontal dimension* of a network complements the vertical dimension
  - ☞ it is not the mere existence of Input-Output connections, but **their demand-supply geometry that critically shapes the transmission of micro-originated shocks**

- Horizontal geometry captured by *network "economic" distances* between sectors
  1. demand-based distances work through production inputs dynamics, while supply-based distances induce opposite comovement due to relative prices and revenue reallocation;
  2. role on sectoral comovement of employment and aggregate output;
     - powerful and independent refinement to prevailing network analysis.

- Comprehensive framework to build distance matrices and form theoretical insights
  - test and corroborate theory-based predictions on sectoral U.S. Input-Output data;
  - suitable for analysis involving the network structure and to design policy prescriptions.

Appendix

Real world example: HBM4 memory chips

- *February 2, 2026* $\leftrightarrow$ Samsung Electronics becomes more technically sophisticated in producing more performative HBM4 memory chips ▸ Barron ▸ investing.com ▸ MarketWatch
  - anticipated production for the Nvidia's AI processor, buying also from Micron;
  - Micron's stock price declines despite solid fundamentals, while Samsung's increases.

- Network-based mechanism through *common downstream buyer*
  - Samsung and Micron $\rightarrow$ slightly different chips but same customer, Nvidia;
  - positive supply shock to Samsung $\rightarrow$ earlier and better production;
  - investors expect lower margins for Micron $\rightarrow$ future downstream demand reallocation.

☞ Horizontal transmission through Input-Output proximity
  - *supply-based* propagation of an idiosyncratic shock without direct network linkages

▸ Go back

Real world example: assembling the AirPods

- *November 2022* $\leftrightarrow$ Goertek loses major Apple AirPods Pro 2 assembly contract, which is reassigned to Luxshare Precision ▸ Yicai Global ▸ Euronews ▸ MacRumors
  - Apple reallocates orders due to production/quality issues at Goertek for AirPods Pro 2;
  - Luxshare gains additional orders and revenue from the same Apple demand, while Goertek experiences a significant revenue reduction (CNY 3.3 billion, USD 455 million).

- Network-based mechanism through *common downstream buyer*
  - Goertek and Luxshare $\rightarrow$ both supply components to Apple;
  - negative demand shock in Goertek $\rightarrow$ Apple reallocates orders to Luxshare;
  - propagation through demand $\rightarrow$ Luxshare benefits, Goertek loses.

☞ Horizontal transmission through Input-Output proximity
  - *supply-based* propagation of an idiosyncratic shock without direct network linkages

▸ Go back

Real world example: OLED panels

- *January 2026* $\leftrightarrow$ Apple reallocates iPhone OLED panel orders from BOE to Samsung Display due to production/quality issues at BOE ▸ MacRumors ▸ TechNode ▸ DisplayDaily
  - BOE faces persistent production/quality issues and Apple cannot meet launch timelines;
  - Apple reallocates millions of OLED panel orders to Samsung Display, which gains additional business, while BOE loses revenues and market share.

- Network-based mechanism through *common downstream buyer*
  - BOE and Samsung Display $\rightarrow$ both supply components (OLED panels) to Apple;
  - negative supply shock in BOE $\rightarrow$ troubles in production efficiency to delivering orders;
  - Apple reallocates its demand $\rightarrow$ Samsung Display benefits, BOE loses.

☞ Horizontal transmission through Input-Output proximity
  - *supply-based* propagation of an idiosyncratic shock without direct network linkages

▸ Go back

Real world example: memory chip suppliers

- *2024-2027* $\leftrightarrow$ surge in demand for DRAM and HBM memory chips driven by AI and cloud data centres
  investing.com · Business Standard · Reuters
  - Samsung, SK Hynix, Micron and other memory suppliers prioritize AI and cloud orders;
  - supplier capacity is limited, causing higher prices and longer lead times;
  - other sectors (*e.g.*, automotive, PC, smartphones) sharing the same suppliers experience indirect supply constraints and cost increases.
- Network-based mechanism through *common upstream seller*
  - AI/cloud workloads and other sectors $\rightarrow$ shared memory suppliers;
  - idiosyncratic demand shock in AI/cloud $\rightarrow$ capacity reallocation at suppliers;
  - increase in upstream price $\rightarrow$ other sectors face indirect constraints and cost pressures.

☞ Horizontal transmission through Input-Output proximity

- *demand-based* propagation of an idiosyncratic shock without direct vertical linkages

Go back

Real world example: cloud-based services

- *2023-2024* $\leftrightarrow$ surge in AI demand increases usage of cloud-based GPU capacity provided by AWS or Microsoft Azure ▸ Times of India ▸ CNBC ▸ Reuters ▸ Financial Times
  - AI users rapidly expand demand for high-end GPUs (*e.g.*, Nvidia H100);
  - cloud providers report capacity constraints and prioritize large AI clients; smaller or non-AI users face higher prices or limited access to GPU instances.

- Network-based mechanism through *common upstream seller*
  - AI and gaming/SaaS/fintech sectors $\rightarrow$ unrelated activities but same cloud provider;
  - demand shock in AI sector $\rightarrow$ providers face GPU capacity constraints;
  - upstream price adjustments $\rightarrow$ other sectors experience cost and quantity effects.

☞ Horizontal transmission through Input-Output proximity
  - *demand-based* propagation of an idiosyncratic shock without direct network linkages

▸ Go back

Real world example: reallocation of TMSC supply

- *January-February 2026* $\leftrightarrow$ surge in AI-chip demand generates unprecedented pressure on TSMC's advanced-node capacity ▸ MacRumors ▸ Business Insider ▸ AppleInsider
  - exceptional demand for AI accelerators increases priority allocation at TSMC;
  - Apple faces tighter access to cutting-edge production nodes.

- Network-based mechanism through *common upstream seller*
  - AI chip producers and Apple $\rightarrow$ different final markets but same foundry (TSMC);
  - positive demand shock in AI sector $\rightarrow$ capacity reallocation at TSMC;
  - reduced effective supply for Apple $\rightarrow$ production inputs potentially from others.

☞ Horizontal transmission through Input-Output proximity
  - *demand-based* propagation of an idiosyncratic shock without direct network linkages

▸ Go back

- The production function is $y_i = z_i\, f^y\left[n_i\ ,\ x_i := \left\{x_{ij}\right\}_{\forall j \in \Phi}\right]$

ASSUMPTION 1 (Production technology requirements)
The production function: (i) has constant returns to scale in either n_i and x_i so that $\alpha_i + \sum_{j\in\Phi} \alpha_{ij} = 1$; (ii) is differentiable, continuous, strictly quasi-concave, homogeneous of degree one, and increasing in z_i, n_i, and x_i; (iii) the case $f^y\left[0\ ,\ \cdot\right]$ is ruled out as labour input is essential to production; (iv) at least two elements of x_i have to be positive, thus $f^y\left[\cdot\ ,\ 0\right]$ cannot exist; consequentially, (v) part of the sectoral output is directly produced by the sector, $x_{ii} > 0$, $\forall i \in \Phi$, that is a production plan with a bundle of intermediate inputs as $f^y\left[\cdot\ ,\ x_{ij}\right]$, given any $j \in \Phi_i$, is precluded.

- note that any functional form chosen for the production function must satisfy the above regularity conditions

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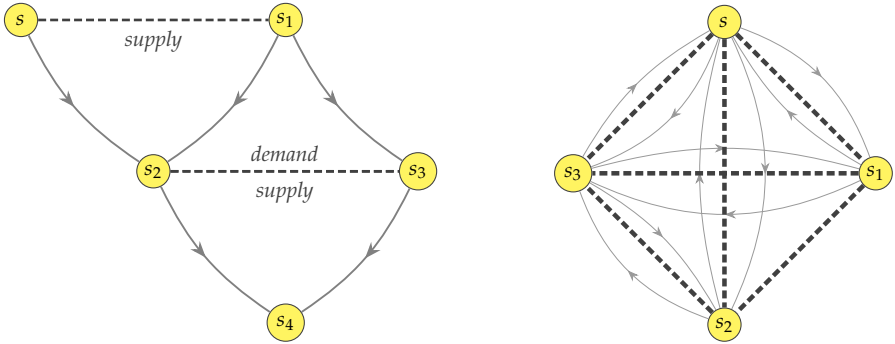
- Consider a pair of sectors, say $\{i, j\}$, related to a common one, say s. Then, for the pair:
  - $\left[\boldsymbol{\mathcal{A}}^2\right]_{ij} = \sum_{s\in\Phi} \alpha_{is}\,\alpha_{sj}$ denotes a recursive production chain;
  - $\left[\boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{A}}'\right]_{ij} = \sum_{s\in\Phi} \alpha_{is}\,\alpha_{js}$ denotes the common upstream neighbourhood;
  - $\left[\boldsymbol{\mathcal{A}}'\boldsymbol{\mathcal{A}}\right]_{ij} = \sum_{s\in\Phi} \alpha_{si}\,\alpha_{sj}$ denotes the common downstream neighbourhood.

Definition 1 (Horizontal proximity and network distance)
The common-supplier and common-buyer proximity matrices are

$$\boldsymbol{\mathcal{B}}^{fd} = \boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{A}}' \qquad \textit{and} \qquad \boldsymbol{\mathcal{B}}^{fs} = \boldsymbol{\mathcal{A}}'\boldsymbol{\mathcal{A}}$$

so that $b^{fd}[i,j] = \langle \boldsymbol{a}_i^{row}, \boldsymbol{a}_j^{row} \rangle$ *and* $b^{fs}[i,j] = \langle \boldsymbol{a}_i^{col}, \boldsymbol{a}_j^{col} \rangle$. *The associated distances are* $d^q[i,j] = \left(b^q[i,i] + b^q[j,j] - 2\,b^q[i,j]\right)^{1/2}$, *for* $q = \{fd, fs\}$, *that is* $d^{fd}[i,j] = \left\|\boldsymbol{a}_i^{row} - \boldsymbol{a}_j^{row}\right\|$ *and* $d^{fs}[i,j] = \left\|\boldsymbol{a}_i^{col} - \boldsymbol{a}_j^{col}\right\|$, *collected in* $\boldsymbol{\mathcal{D}}^q$.

Go back



(a) Distance definition

(b) Distance network

Figure A.1: Horizontal Input-Output geometries

Go back

- *input demand*: consider the case in which two sectors, say $\{i, j\}$, are buying intermediate inputs from the same sector, say s. The comovement induced by the common sector-s is

$$b^{fd}_{s\rightarrow}[i,j] \;=\; \text{Cov}\left[d\log\kappa_{i,s}\,,\, d\log\kappa_{j,s}\right] \;=\; \alpha_{is}\alpha_{js} \;\geq\; 0 \tag{A.1}$$

- *input supply*: consider the case in which two sectors, say $\{i, j\}$, are selling their own commodity as an intermediate input to the same sector, say s. The comovement induced by the common sector-s is

$$b^{fs}_{\rightarrow s}[i,j] \;=\; \text{Cov}\left[d\log\left[p_i x_{si}\right],\, d\log\left[p_j x_{sj}\right]\right] \;=\; \alpha_{si}\alpha_{sj} \;\geq\; 0 \tag{A.2}$$

Go back

LEMMA 1 (Properties)

Demand- and supply-based matrices satisfy:

- $\boldsymbol{\mathcal{B}}^{fd}$ *and* $\boldsymbol{\mathcal{B}}^{fs}$ *are non-negative, symmetric and positive semi-definite;*
- *proximity is coupled with* $d^q[\cdot]$*, a Euclidean metric on Input-Output profiles:* $d^q[i,j] = 0$ *if and only if the two profiles coincide;* $\boldsymbol{\mathcal{D}}^{fd}$ *and* $\boldsymbol{\mathcal{D}}^{fs}$ *have a zero diagonal and, unless null, are not positive semi-definite; for profiles normalized to sum to one at the common node,* $d^q[i,j] \in \left[0, \sqrt{2}\right]$ *for* $q = \{fd, fs\}$*;*
- *proximity and distance matrices make all nodes simultaneously connected, with each cell strictly positive if the pair shares at least one neighbour, so that both matrices are full;*
- *they differ entry-wise unless the corresponding bilateral trade with the common neighbour coincides:* $\boldsymbol{\mathcal{B}}^{fd} = \boldsymbol{\mathcal{B}}^{fs}$ *if and only if* $\boldsymbol{\mathcal{A}}$ *is a normal matrix, as when* $\boldsymbol{\mathcal{A}}$ *is symmetric.*

Go back

Interpretation of matrix $\mathcal{D}$

- On the theoretical side, each cell identifies "network proximity"
  - the higher $b^q[i,j]$, the larger the transmission $\longleftrightarrow$ stronger horizontal complementarity

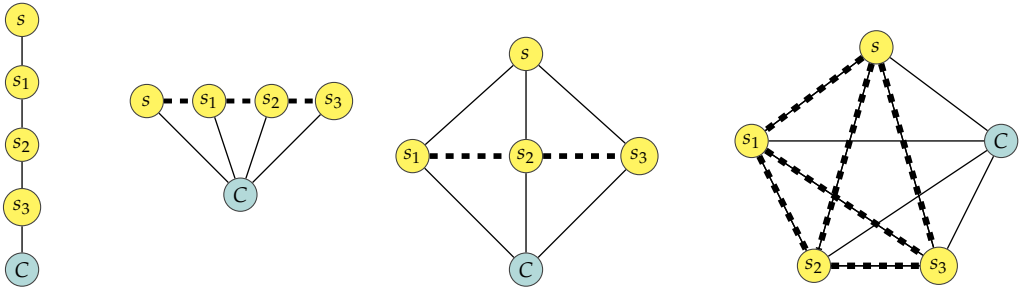
- Corresponding economic distance as a Gram-to-metric conversion. For input demand:

$$d^{fd}[i,j]^2 = b^{fd}[i,i] + b^{fd}[j,j] - 2\,b^{fd}[i,j] = \left\|\boldsymbol{a}_i^{row} - \boldsymbol{a}_j^{row}\right\|_2^2 \tag{A.3}$$

  - *Euclidean distance*, $d^{fd}[i,j] = \sqrt{\left[\alpha_{ii} - \alpha_{ji}\right]^2 + \left[\alpha_{ij} - \alpha_{jj}\right]^2 + \cdots + \left[\alpha_{im} - \alpha_{jm}\right]^2}$;
  - each matrix is *symmetric* outside the main diagonal, with $\text{diag}\left[\boldsymbol{\mathcal{D}}^j\right] = 0$.

- These are the *intensive margin*, with values inside $[0, \infty)$. To account for *extensive margin* (discrete) values, I employ the *shortest path algorithm*
  - that is, how many steps are necessary for a sector to reach another one

Go back



(**a**) Sequential (**b**) Rhomboid (**c**) Star

Figure A.2: Benchmark economies

Go back

Reduced forms for distance-lead comovement

Proposition 1 (Common-supplier transmission)

At first round, the purchases of downstream sector-i from the shared upstream supplier-s respond as

$$\delta_{s\rightarrow}[i,j] = \frac{\partial \log x_{is}}{\partial \log x_{js}} = -\chi_{is}\ \widetilde{\varphi}_s\ e_{sj}$$

with $\chi_{is} = \sigma_i\left(1-\widehat{\alpha}_{is}\right) + \alpha_i\,\widehat{\alpha}_{is} + \eta_i\,\alpha_{is} > 0$ *and* $\widetilde{\varphi}_s = \varphi_s \big/ \left[1+\varphi_s\left(e_{s0}+\sum_{i\in\Phi} e_{si}\chi_{is}\right)\right]$. *Sector-i's output responds as* $\frac{\partial \log y_i}{\partial \log x_{js}} = -\eta_i\,\lambda_j \sum_{s\in\Phi}\alpha_{is}\frac{\widetilde{\varphi}_s}{\lambda_s}\,\alpha_{js}$.

Proposition 2 (Common-buyer transmission)

At first round, the sales of upstream sector-i to the shared downstream buyer-s respond as

$$\delta_{\rightarrow s}[i,j] = \frac{\partial \log\left[p_i\,x_{si}\right]}{\partial \log p_j} = \left[\,(\sigma_s-1)+(1-\alpha_s)\,(1-\eta_s)\,\right]\widehat{\alpha}_{sj}$$

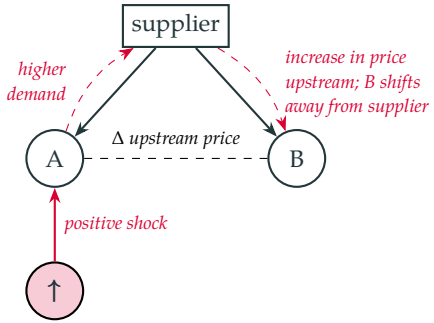
and, summing over the common buyers $\forall s \neq i,j$, *the corresponding component of i's total sales responds as* $\frac{\partial \log p_i y_i}{\partial \log p_j} = \frac{1}{\lambda_i}\sum_{s\in\Phi_{i,j}}\lambda_s\left[\,(\sigma_s-1)+(1-\alpha_s)\,(1-\eta_s)\,\right]\alpha_{si}\,\widehat{\alpha}_{sj}$.

▸ Go back

▸ Graphs

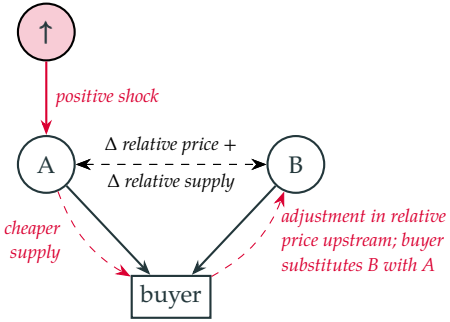
Graphing the economics of distance-lead comovement

Demand distance: upstream pass-through



downstream shifting away from upstream input
⇒ vertical price effect + horizontal ambiguity downstream

Supply distance: downstream pass-through



upstream inputs used jointly downstream
⇒ demand reallocation and clear horizontal transmission

▸ Go back

▸ Intuition ▸ Interpretation

Economic interpretation of reduced-forms

- Intuition beyond the horizontal transmission:
  - *demand distance*: shocks trigger downstream buyer to shift away from upstream suppliers
    - ⋆ vertical and horizontal effects get mixed, making comovement ambiguous to interpret
  - *supply distance*: upstream shocks pass downstream, where inputs are used together
    - ⋆ horizontal propagation among suppliers is direct, mechanical, and unambiguous
- Core contrast:
  - *demand distance*: mere variation in upstream price (no true horizontal complementarities);
  - *supply distance*: substitution of upstream inputs through downstream demand reshuffling.
- Downstream complementarities (*i.e.*, supply distances) matter the most to generate opposite comovement {in line with Atalay (2017) and Baqaee and Farhi (2019)}

Go back

Depth interpretation of reduced-forms

- ***Demand-based network distances***
  - capture how easily a downstream buyer can switch among suppliers, without creating "true" horizontal linkages $\longleftrightarrow$ *upstream pass-through effect*;
  - reflect substitution patterns across buyers sharing common upstream suppliers;
  - ambiguous propagation effects, as they lack genuine complementarities among buyers.

- ***Supply-based network distances***
  - generate genuine horizontal complementarities among sectors whose outputs are jointly used by common downstream buyers $\longleftrightarrow$ *downstream pass-through effect*.
  - shocks to one supplier propagate horizontally as other suppliers adjust production due to downstream complementarities;
  - strong comovement effects, consistent with non-linear network theories emphasizing complementarities in intermediate inputs. {*e.g.*, Atalay (2017), Baqaee and Farhi (2019)}

Go back

Theorem 1: widespread horizontal geometry

Theorem 1 (On the vertical-horizontal factorization)

To first order, Domar weights respond to sectoral shocks as

$$d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}' \boldsymbol{\mathcal{H}} \boldsymbol{\mathcal{L}}\, d\log \mathbf{z} \qquad \text{with} \quad \boldsymbol{\mathcal{H}} = \sum_{j \in \Phi} (\sigma_j - 1)\, \lambda_j\, (1 - \alpha_j)\, \boldsymbol{\mathcal{C}}_j$$

Equivalently, $\frac{\partial \lambda_m}{\partial \log z_i} = \sum_{j \in \Phi} (\sigma_j - 1)\, \lambda_j\, (1 - \alpha_j)\, \mathrm{Cov}_{\widehat{\boldsymbol{a}}_j^{row}} \left[\boldsymbol{\ell}_i^{col}, \boldsymbol{\ell}_m^{col} \right]$*, where* $\boldsymbol{\ell}_i^{col}$ *is the i-th column of* $\boldsymbol{\mathcal{L}}$ *and the covariance is across buyer-j's intermediate inputs with probabilities* $\widehat{\boldsymbol{a}}_j^{row}$*.*

Corollary 1 (Signs)

Let sector-s to be the shared node of sectors $\{i, j\}$*. Then:*

- *under Cobb-Douglas intermediates* ($\sigma_s = 1$) *it holds* $\boldsymbol{\mathcal{H}} = \mathbf{0}$*;*
- *if* $\sigma_s \geq 1$ ($\sigma_s \leq 1$)*,* $\forall s \in \Phi$*, then* $\boldsymbol{\mathcal{H}}$ *and* $\boldsymbol{\mathcal{R}}$ *are positive* (*negative*) *semi-definite, and all off-diagonal entries of* $\boldsymbol{\mathcal{H}}$ *are non-positive* (*non-negative*)*, and first-tier co-suppliers are rivals* (*complements*) *in sales.*

Go back

Further corollaries for Theorem 1

Corollary 2 (Domar weights do not span horizontal geometry)

There exist economies with identical elasticities and identical Domar weights – hence identical first-order responses of aggregate output to any vector of shocks – in which the same pair of sectors are rivals in one economy and unrelated in the other.

Corollary 3 (Horizontal proliferation)

It holds $\mathcal{H}_{i,j} \neq 0$ if and only if $b^{fs}[i,j] > 0$, and:

- *if buyer-s starts purchasing from supplier-i, $\boldsymbol{\mathcal{A}}$ gains one positive entry, α_{si}, while $\mathcal{H}_{i,j}$ gains a non-zero term for every $j \in \Phi_i$ with $\alpha_{sj} > 0$;*
- *if buyer-s stops purchasing from supplier-i, $\boldsymbol{\mathcal{A}}$ loses one positive entry, while $\mathcal{H}_{i,j}$ vanishes only for the suppliers $j \in \Phi_i$ whose only common buyer with i was s;*
- *relative to buyer-s's intermediate purchases, $\lambda_s(1-\alpha_s)$, the rivalry that s generates in $\boldsymbol{\mathcal{H}}$ is $|\sigma_s - 1|\left(1 - \|\widehat{\boldsymbol{a}}_s^{row}\|^2\right)$, where $\|\widehat{\boldsymbol{a}}_s^{row}\|^2$ is the Herfindahl index of s's purchases.*

Go back

- How does sectoral employment fluctuate? $\longleftrightarrow$ ***vertical*** + ***horizontal*** dimensions

PROPOSITION 3 (Employment reallocation)
To first order, sectoral employment responds as

$$d \log \boldsymbol{n} = d \log N \, \mathbf{1} + \boldsymbol{\Lambda}^{-1} \boldsymbol{\mathcal{R}} \, d \log \boldsymbol{z} \qquad \textit{with} \quad \boldsymbol{\Lambda} = \text{Diag}\,[\boldsymbol{\lambda}]$$

Moreover, $\boldsymbol{\alpha}' \boldsymbol{\mathcal{R}} = \mathbf{0}$*: the employment-weighted sum of relative responses is zero.*

  - *common component* $\rightarrow$ same shifts all sectors, no information on the network;
  - *reallocation component* $\rightarrow$ change in sales shares maps into a change in employment

- Two predictions:
  - a shock that moves the employment of a sector moves that of its first-tier common- buyer neighbours in the opposite direction;
  - horizontally distant sectors (small $\boldsymbol{\mathcal{R}}$) comove only through the common component.

THEOREM 2 (On horizontal geometry and aggregate output)

The impact of the network's horizontal structure on aggregate GDP is as follows:

- *to first order,* $d \log Y = \boldsymbol{\lambda}' \boldsymbol{\varepsilon}$*, so that* $\operatorname{Var}[d \log Y] = \nu^2 \|\boldsymbol{\lambda}\|^2$ *does not depend on* $\boldsymbol{\mathcal{H}}$ *and vanishes as* $m \to \infty$ *when* $\lambda_i \asymp \frac{1}{m}$*;*
- *to second order,* $d^2 \log Y = \boldsymbol{\varepsilon}' \boldsymbol{\mathcal{L}}' \boldsymbol{\mathcal{H}} \boldsymbol{\mathcal{L}} \boldsymbol{\varepsilon} = \sum_{j \in \Phi} (\sigma_j - 1) \lambda_j (1 - \alpha_j) \operatorname{Var}_{\widetilde{\boldsymbol{a}}_j^{row}} [\boldsymbol{\mathcal{L}} \boldsymbol{\varepsilon}]$*;*
- *consequently, it holds* $\mathbb{E}[\log Y] - \log Y^{det} = \frac{1}{2} \nu^2 \operatorname{Trace}[\boldsymbol{\mathcal{L}}' \boldsymbol{\mathcal{H}} \boldsymbol{\mathcal{L}}] + o[\nu^2]$.

Go back

Production network data

- ***market share, after redefinitions*** Input-Output tables
  - square matrix, $\underset{m \times m}{\boldsymbol{X}}$, whose diagonal correspond to sector-share of own production while:
    - $\star$ $\underset{m \times m}{\boldsymbol{X}^{up}}$ = rows $[\boldsymbol{X}]$ is the share of production of sector-i that is bought by sector-j;
    - $\star$ $\underset{m \times m}{\boldsymbol{X}^{dw}}$ = columns $[\boldsymbol{X}]$ is the share of production that sector-i buys from sector-j.
  - in the dataset built, I compute $\boldsymbol{x} = \mathbf{1} - \text{diag}\,[\boldsymbol{X}]$ to extract the *total share* of production that sectors buy from other sectors, where $\mathbf{1}$ an $m \times 1$ vector of ones

- ***commodity-by-commodity total direct requirements*** Input-Output tables
  - square matrix, $\underset{m \times m}{\boldsymbol{\mathcal{A}}}$, in which each entry corresponds to:
    - $\star$ $\underset{m \times m}{\boldsymbol{\mathcal{A}}^{up}}$ = rows $[\boldsymbol{\mathcal{A}}]$ is the dollar value of one unit of sector-s that is bought by sector-s';
    - $\star$ $\underset{m \times m}{\boldsymbol{\mathcal{A}}^{dw}}$ = columns $[\boldsymbol{\mathcal{A}}]$ is the dollar value of sector-s in order to buy one unit from sector-s'

→ *row sectors* identify *supplier* (*upstream*), and *column sectors* identify *buyer* (*downstream*)

Go back

Interpretation, cells

Interpreting an Input-Output matrix, cells

	AGRICULTURE	MANUFACTURING	SERVICES
AGRICULTURE		AGRICULTURE **SELLS** TO MANUFACTURING	AGRICULTURE **SELLS** TO SERVICES
MANUFACTURING	AGRICULTURE **BUYS** FROM MANUFACTURING		MANUFACTURING **SELLS** TO SERVICES
SERVICES	AGRICULTURE **BUYS** FROM SERVICES	MANUFACTURING **BUYS** FROM SERVICES	

Go back

Production network matrix

- To investigate the network structure:
  - → upstream and downstream sectors, as well as sectoral distances, are computed through the *commodity-by-commodity direct requirements* table as of 2007;
  - → only to compute the Leontief inverse, disregard small transactions

$$\alpha^{new}\left(s,s'\right)=\frac{\alpha\left(s,s'\right)}{\alpha\left(s\right)}\quad \textit{is set to}\quad 0\quad \textit{if}\quad \alpha^{new}\left(s,s'\right)\leq 0.01$$

  where $\alpha\left(s\right)=\sum_{s'\in row}\alpha\left(s,s'\right)$ or $\alpha\left(s\right)=\sum_{s'\in column}\alpha\left(s,s'\right)$

- Defining $\boldsymbol{\alpha}=\left[\alpha\left(s\right),\alpha\left(s'\right),\ldots,\alpha\left(m\right)\right]'$, the considered Input-Output structure is

$$\underset{m\times m}{\boldsymbol{\mathcal{A}}^{new}}=\underset{m\times m}{\boldsymbol{\mathcal{A}}'}\underset{m\times 1}{\boldsymbol{\alpha}^{-1}}$$

  in which each cell, $\alpha^{new}\left(s,s'\right)$, is greater than 1%

Go back

Network representation

share of sectoral total sales

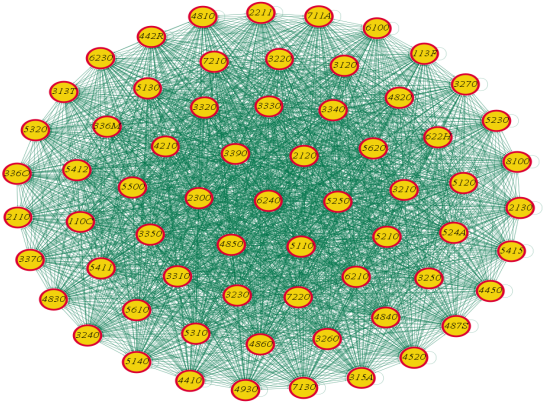


Figure B.1: Production network of the U.S. economy in 2007

Go back

Network representation

share of sectoral total purchases

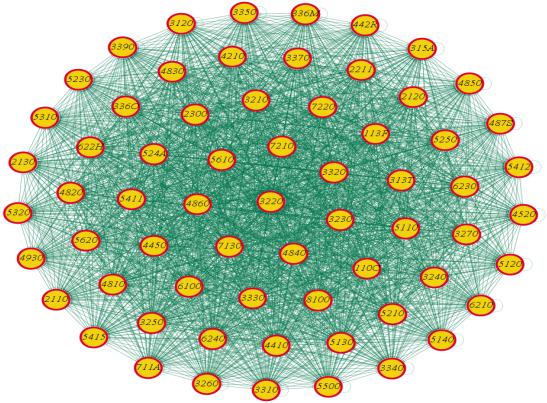


Figure B.2: Production network of the U.S. economy in 2007

Go back

Intensive margin values for network distance

- System of *Cartesian coordinates* to compute the *network distances* {*e.g.*, Conley and Dupor (2003)}:
  1. ***factor input demand***, when buying similar inputs. Given $\alpha_{ij}^{fd} = \frac{\alpha_{ij}}{\sum_{j\in\Phi}\alpha_{ij}}$, then

$$d^{fd}[i,j] = \left\{ \sum_{s\in\Phi} \left[\alpha_{is}^{fd} - \alpha_{js}^{fd} \right]^2 \right\}^{\frac{1}{2}}$$

     is measuring downstream proximity of production function requirements. ▸ Heatmap, demand
  2. ***factor input supply***, when selling to similar sectors. Given $\alpha_{ij}^{fs} = \frac{\alpha_{ij}}{\sum_{i\in\Phi}\alpha_{ij}}$, then

$$d^{fs}[i,j] = \left\{ \sum_{s\in\Phi} \left[\alpha_{si}^{fs} - \alpha_{sj}^{fs} \right]^2 \right\}^{\frac{1}{2}}$$

     is measuring upstream comovement of sectoral revenues from trade; ▸ Heatmap, supply
- Distances are computed as those from the *Total Requirements* ▸ Distributions ▸ Correlation
  - two sectors are close when they share partners both locally and throughout the depth of their supply chains

▸ Go back

Sectoral distance in the network, factor input demand

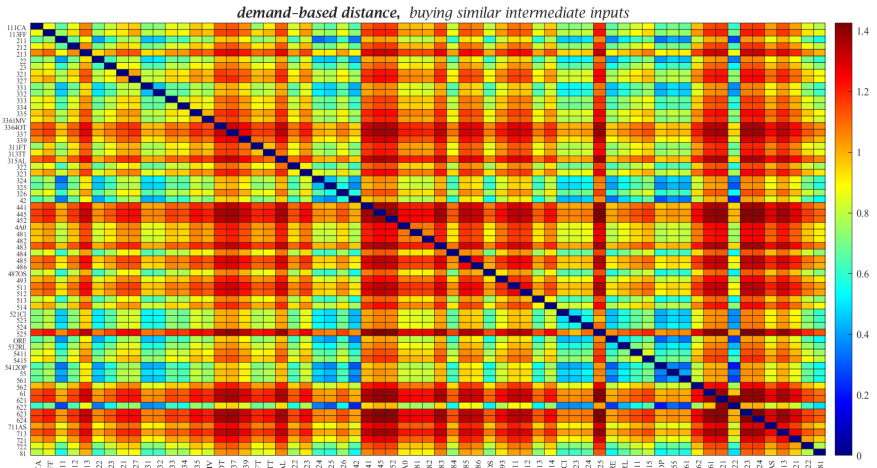


Figure B.3: *Factor demand distances* in the Input-Output matrix, U.S. 2007

Go back

Sectoral distance in the network, factor input supply

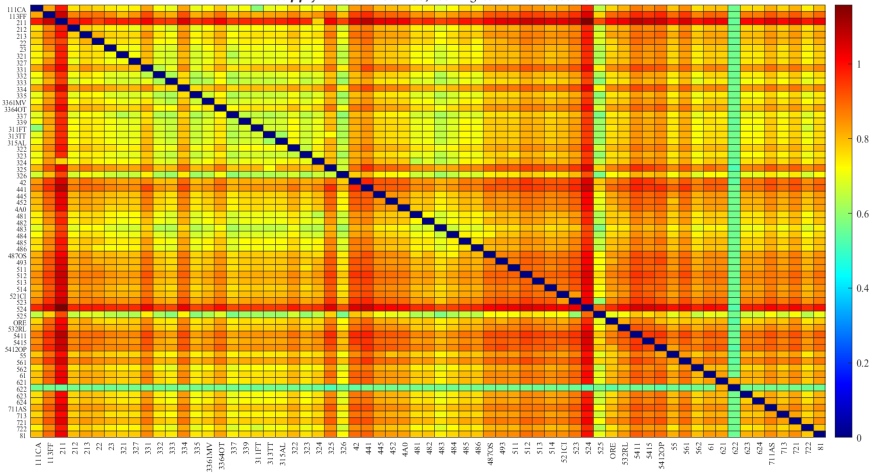
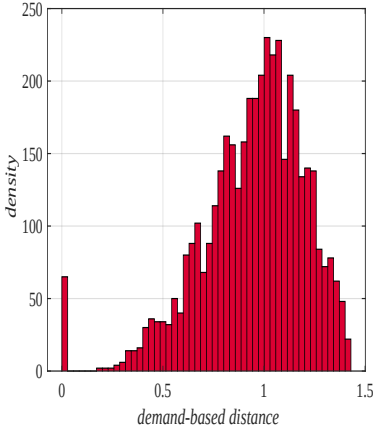


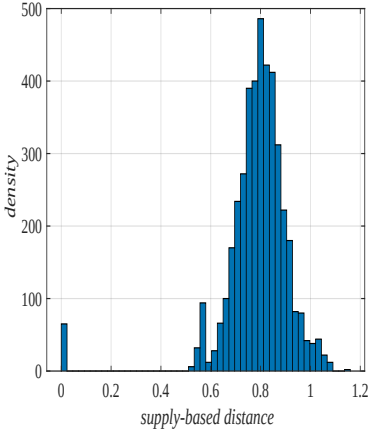
Figure B.4: *Factor supply distances* in the Input-Output matrix, U.S. 2007

Go back

Distribution of intensive margin network distances



(**a**) Factor input demand



(**b**) Factor input supply

Figure B.5: Distribution of network economic distances

Go back

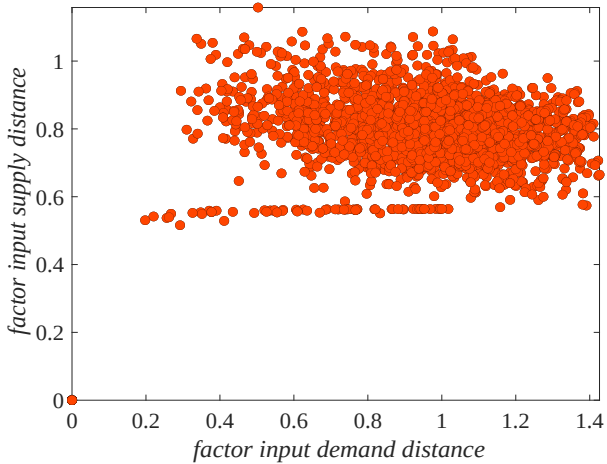


Figure B.6: Correlation of network economic distances

Go back

Shortest path algorithm.

Central idea is to find, for each single node, all the other nodes that can be reached in one step, then in two steps, and so on. Each time a node is reached for the first time, it records the number of steps it took to get there — this is the shortest path approach. In other words, find the shortest number of steps it takes to get from every node to every other node in a network. In particular:

(*a*) identify the direct connections between any pair of nodes (those that take exactly one step to be connected);

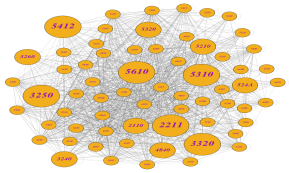
(*b*) for not-connected pairs, increase the path length by 1 in each round (two steps, then three steps, etc.). Multiply the graph by itself to discover which new pairs of nodes are now connected through longer paths;

(*c*) if a pair of nodes becomes connected (*i.e.*, only if it has not be already found a shorter path), record the current length as the shortest distance between them;

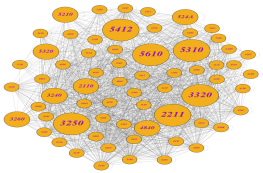
(*d*) repeat this process until any new reachable pairs of nodes is found. For any pair of nodes that are still unreachable (*i.e.*, the distance is still zero and they are not linked in the network), set their distance to infinity.

Go back

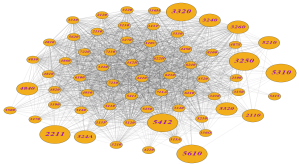
□



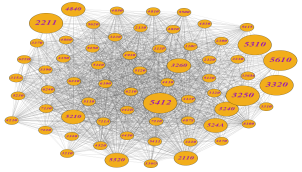
(**a**) Demand, $d = 1$



(**b**) Supply, $d = 1$



(**c**) Demand, $d = 2$



(**d**) Supply, $d = 2$

Figure B.7: Visualization of production network distances

Go back

Characteristics of network-based distances, extensive margin

(1) The distribution of horizontal complementarities is uneven

- the factor input demand network is slightly denser than the supply network at $d = 1$, but then this asymmetry vanishes ($d = 2$)

(2) Sectors with the major Input-Output linkages tend to concentrate around the nucleus

- the most connected hubs (top 10%) and the mid-hubs (top 20%) are similarly central

(3) Sectors with the highest number of production network connections tend to be evenly central at shorter distances, and relatively more peripheral at longer distances

☞ The structure of horizontal complementarities depend on the notion of distance

- the same vertical production system generates distinct horizontal geometries

Go back

Table B.1: Dimensional dominance

	network sparsity				
	15%	30%	40%	50%	95%
	$\tau = 0.5\%$	$\tau = 1\%$	$\tau = 1.5\%$	$\tau = 2\%$	$\tau = 5\%$
vertical propagation	.2133	.1995	.1803	.1497	.0762
horizontal propagation	.1371	.1559	.1718	.1731	.1746

Spectral norms for a 3-digit U.S. Input-Output network in year 2007 when small transactions, at different threshold-τ, are removed. Source: BEA and own calculations.

Go back

Network exposures and aggregate shocks: interpretation

① ***Aggregate TFP shock***

- *supply-based* horizontal exposure: large, positive, and *persistent*
  - ⋆ sectors sharing common downstream buyers react more strongly to TFP innovations
- *demand-based* horizontal exposure: statistically insignificant at all horizons
- *vertical* (Leontief inverse) exposure: positive but smaller, quickly dissipating

☞ aggregate productivity diffuses primarily through *common downstream demand* linkages, not through observed Input-Output chains

② ***Government spending shock***

- *demand-based* horizontal exposure: positive, significant, and *persistent*
  - ⋆ sectors with common suppliers to government-exposed industries amplify fiscal shocks
- *supply-based* horizontal exposure: negative, significant, and *persistent*
  - ⋆ weaker responses, consistent with competitive demand reallocation
- *vertical* (Leontief inverse) exposure: statistically insignificant at all horizons

☞ horizontal geometries are the *primary* propagation channel for fiscal disturbances; dominant channel varies across shock types

‣ Go back

Defining independent variables

- sectoral output and employment growth, $\dot{var}_{i,t} = var_{i,t} - var_{i,t-1}$, $var = \{y, n\}$;
- network distance-d computed as a *shortest path problem*, unweighted (Corollary 2);
- given that sectors are $l \in \{upstream, downstream\}$ to sector-i:
  - (a) distance-d growth, $\dot{var}_t(\Phi_i, d) = \sum_{j \neq i} \left[var_{j,t}(d) - var_{j,t-1}(d) \right]$;
  - (b) layer type-l growth, $\dot{var}_t(\Phi_i, l) = \sum_{j \neq i} \ell_{i,j}(l) \left[var_{j,t}(l) - var_{j,t-1}(l) \right]$;
- controls:
  - (a) sectoral employment size, n_i, value-added y_i, and their growth rates, and net exports;
  - (b) aggregate employment size, N, aggregate value-added, Y, and their growth rates.

Go back, employment · Go back, I-O shares

Empirical framework: identification step

① Purge the dynamics of sectoral I-O shares from movements in sectoral employment

- panel Fixed-Effects (FE) regression over $s \in \Phi$ sectors

$$x_{i,t} = \beta_x^{cons} + \underbrace{\beta_n \, \dot{n}_{i,t}}_{\text{emp. growth}} + \underbrace{\sum_d \beta_{n(d)} \, \dot{n}_t(\Phi_i, d)}_{\text{emp. growth, network distance}} + \underbrace{\sum_l \beta_{n(l)} \, \dot{n}_t(\Phi_i, l)}_{\text{emp. growth, upstream and downstream}}$$

$$+ \underbrace{\gamma_i' \, \mathcal{Z}_{i,t} + \Gamma_i' \, \dot{\mathcal{Z}}_{i,t}}_{\text{sector-level controls}} + \underbrace{\gamma' \, \mathcal{Z}_t + \Gamma' \, \dot{\mathcal{Z}}_t}_{\text{aggregate controls}} + \underbrace{\phi_i}_{\text{sector Fe}} + u_{i,t}^x$$

- estimated residual $\widehat{u}_{i,t}^x$ identify the residual variation in I-O shares

▸ Variables' definition

② Estimate panel Local Projections (LP) to study the dynamic effects of changes in sectoral I-O intermediate shares on its (cumulative) employment

- given $h = 5$ horizons, sequence of panel predictive regressions of the form

$$\dot{n}^{cum}_{i,t+h} = \beta_{u^x,h}\, \widehat{u}^x_{i,t} + \underbrace{\phi_{t+h} + \phi_i}_{\textit{horizon and sector Fe}} + \nu^x_{i,t+h}$$

... also considering d^{th}-distance across sectors

$$\dot{n}^{cum}_{i,t+h} = \beta_{u^x_j,h}\,(d) \sum_{j \in \Phi_i} \widehat{u}^x_{j,t}\,(d) + \underbrace{\phi_{t+h} + \phi_i}_{\textit{horizon and sector Fe}} + \nu^x_{i,t+h}\,(d)$$

- variables at h identifies the cumulative difference between time t and $t + h$

Network shares and employment

factor input demand and directed network

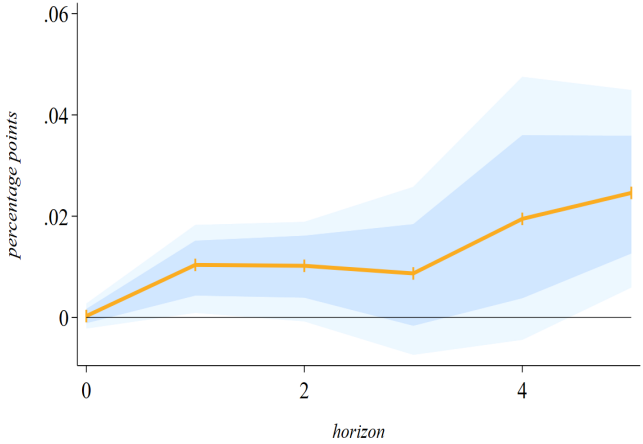
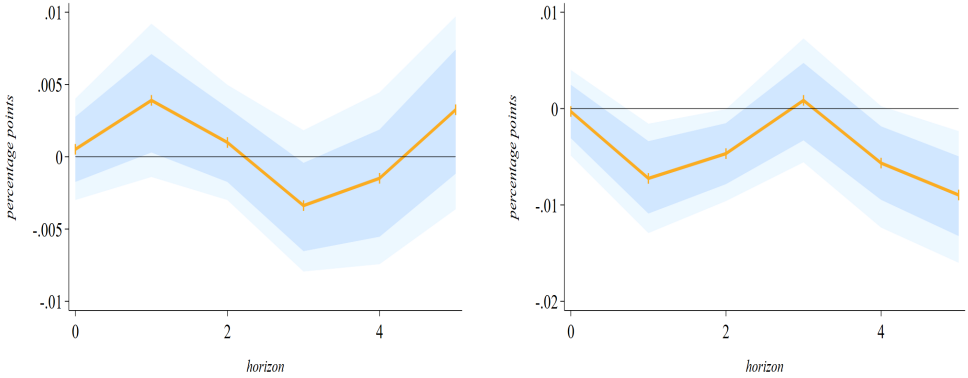


Figure D.1: Sectoral employment response following a 1% increase in own I-O share

Go back

Network shares and employment, distance

factor input demand and directed network



(**a**) $d = 1$ (**b**) $d = 2$

Figure D.2: Sectoral employment response following a 1% increase in I-O share of nearby sectors

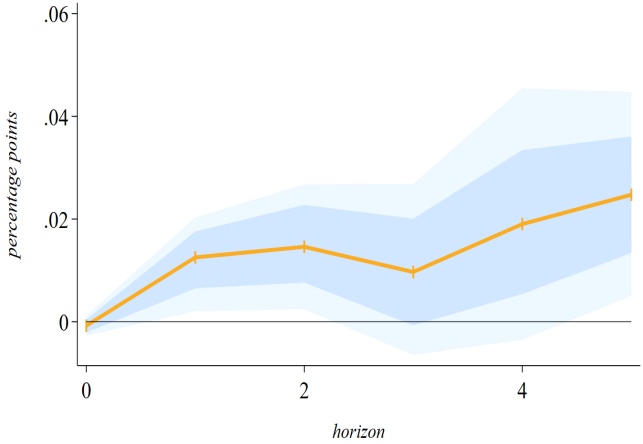
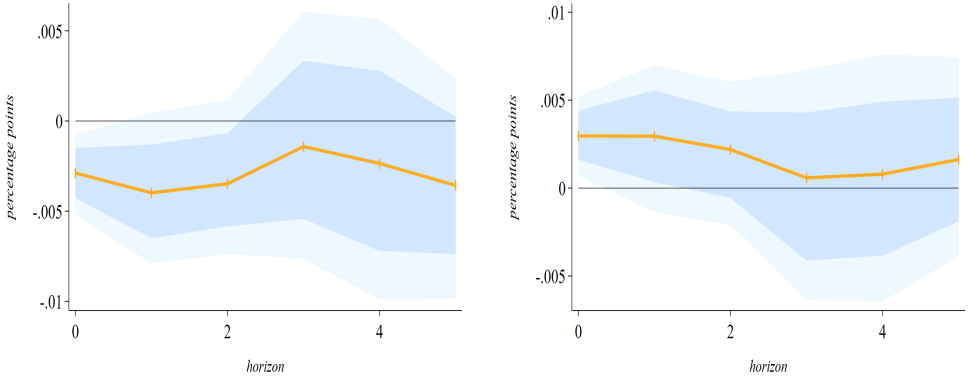


Figure D.3: Sectoral employment response following a 1% increase in own I-O share

Go back

Network shares and employment, distance

factor input supply and directed network



(**a**) $d = 1$ (**b**) $d = 2$

Figure D.4: Sectoral employment response following a 1% increase in I-O share of nearby sectors

Sectoral comovement of employment

① Identification, to extract exogenous dynamics {as in Barattieri and Cacciatore (2023)}

$$n_{i,t} = \beta_n^{cons} + \underbrace{\beta_y \, \dot{y}_{i,t}}_{\text{output growth}} + \underbrace{\sum_d \beta_{y(d)} \, \dot{y}_t(\Phi_i, d)}_{\text{output growth, network distance}} + \underbrace{\sum_l \beta_{y(l)} \, \dot{y}_t(\Phi_i, l)}_{\text{output growth, upstream and downstream}}$$

$$+ \underbrace{\gamma_i' \mathcal{Z}_{i,t} + \Gamma_i' \dot{\mathcal{Z}}_{i,t}}_{\text{sector-level controls}} + \underbrace{\gamma' \mathcal{Z}_t + \Gamma' \dot{\mathcal{Z}}_t}_{\text{aggregate controls}} + \underbrace{\phi_i}_{\text{sector Fe}} + u_{i,t}^n$$

- estimated residual $\widehat{u}_{i,t}^n$ identifies the residual variation in sectoral employment

② Local Projections, for short-run dynamic responses {as in Barattieri and Cacciatore (2023)}

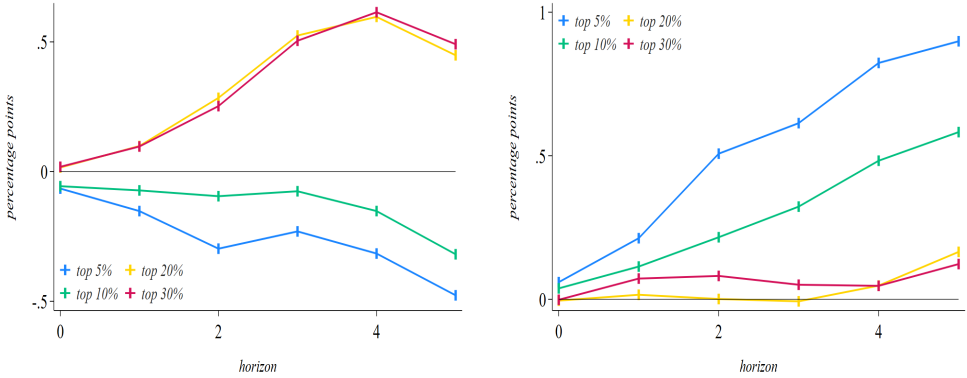
$$\dot{n}_{i,t+h}^{cum} = \sum_{j \in \Phi_i} \beta_{u_j^n, h}(d) \; \widehat{u}_{j,t}^n(d) + \underbrace{\phi_{t+h} + \phi_i}_{\text{horizon and sector Fe}} + \nu_{i,t+h}^n(d)$$

▸ Variables' definition

Employment effects of most connected sectors, distance

Overview

factor input demand and directed network



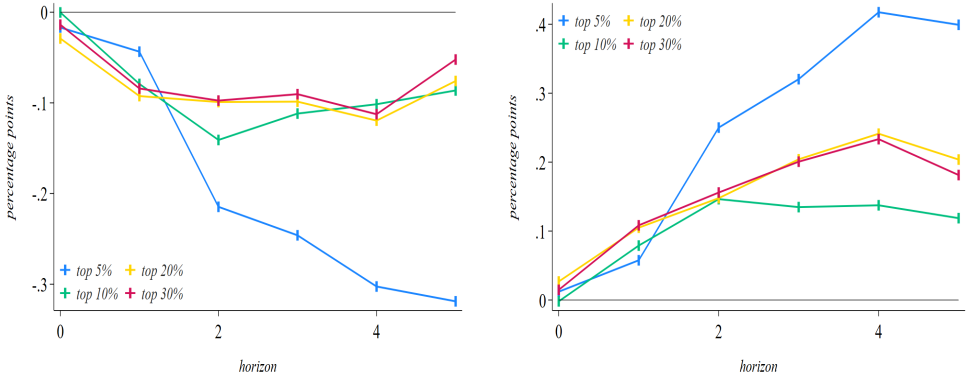
(**a**) $d = 1$, *directed* (**b**) $d = 2$, *directed*

Figure E.1: Demand linkages and comovement in interlinked sectors

Average, directed, demand

Employment effects of most connected sectors, distance

factor input supply and directed network



(**a**) $d = 1$, *directed* (**b**) $d = 2$, *directed*

Figure E.2: Supply linkages and comovement in interlinked sectors

Average, directed, supply

- ***peripheral sectors*** ▸ Peripheral sectors
  - *intuition*: a shock in a central sector transmits outward to sectors "at the margins" of the network, inducing broader fluctuations;
    - ⋆ if network distances were irrelevant, comovement would be expected universally;
  - *results*: opposite comovement under both minor demand and supply distances; positive for shocks originated in further sectors.

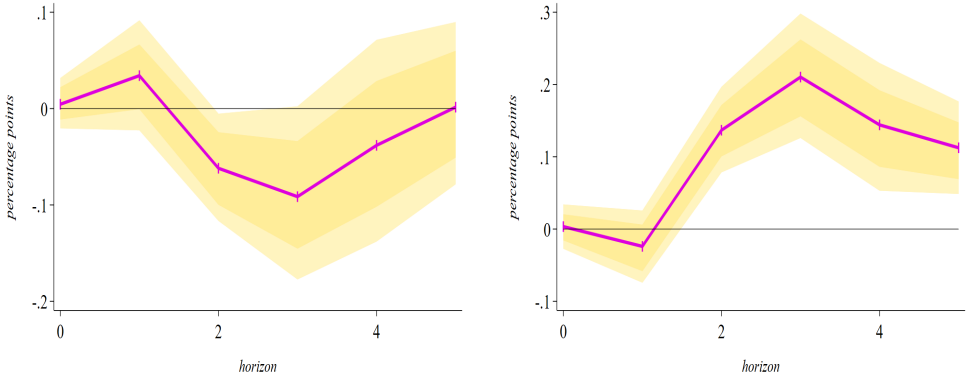
- ***inventories*** ▸ Changes in private inventories
  - *intuition*: inter-temporal buffer that smooths production responses, so that the contemporaneous nature of horizontal complementarities is attenuated
  - *results*: horizontal transmission vanishes when holding inventories; in contexts without inventories, the baseline results are preserved

- ***different base year*** *in the paper*
  - *intuition*: the production network structure exhibits limited evolution over time;
  - *results*: robust if replicated for alternative benchmark years (2002 and 2012).

Employment effects, distance

factor input demand and directed network

Overview



(a) $d = 1$ **(b)** $d = 2$

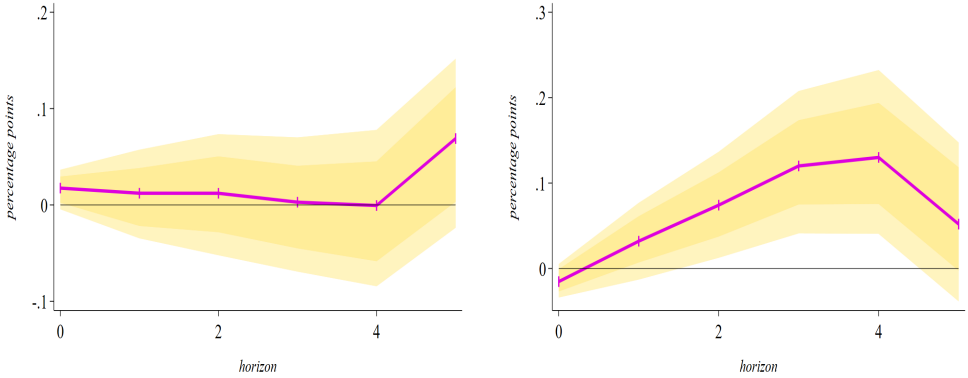
Figure E.3: Employment comovement under direct demand linkages

Go back

Employment effects, distance

factor input supply and directed network

Overview



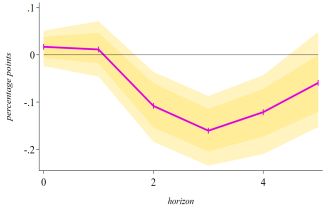
(a) $d = 1$ **(b)** $d = 2$

Figure E.4: Employment comovement under direct supply linkages

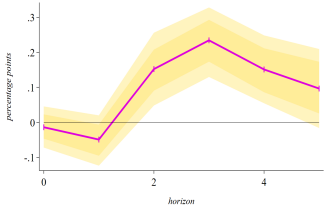
Go back

Employment effects in most peripheral sectors, distance

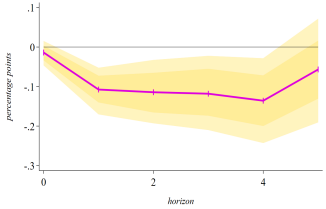
factor input demand, factor input supply and directed network



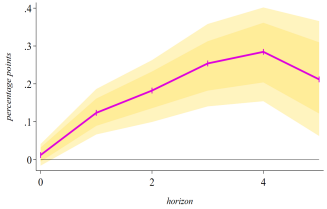
(**a**) $d = 1$, demand



(**b**) $d = 2$, demand



(**c**) $d = 1$, supply

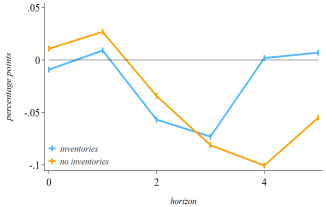


(**d**) $d = 2$, supply

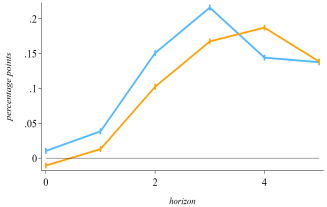
Go back

Employment effects given changes in private inventories, distance c

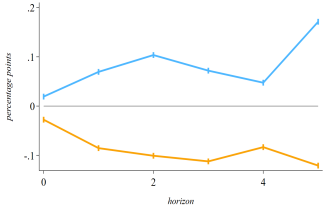
factor input demand, factor input supply and directed network



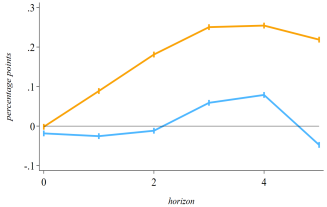
(**a**) $d = 1$, demand



(**b**) $d = 2$, demand



(**c**) $d = 1$, supply



(**d**) $d = 2$, supply

Go back